

TEST OF MATHEMATICS FOR UNIVERSITY ADMISSION

PAPER 1: APPLICATIONS OF MATHEMATICAL KNOWLEDGE

Practice paper · Set 11

75 minutes

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but do not open the question paper until you are told that you may do so.

A separate answer sheet is provided for this paper. You require a soft pencil and an eraser. Please complete the answer sheet with your candidate number, centre number, date of birth and full name.

This paper contains **20 questions**. For each question, choose the **one** answer you consider correct and record your choice on the separate answer sheet. If you make a mistake, erase thoroughly and try again.

There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all 20 questions. Each question is worth one mark.

You can use the question paper for rough working or notes, but no extra paper is allowed.

You must complete the answer sheet within the time limit.

Calculators and dictionaries are NOT permitted.

There is no formulae booklet for this test.

Please wait to be told you may begin before turning this page.

This is a practice paper generated from a question bank written to the TMUA content specification. It is not an official UAT-UK paper. Answers and worked solutions are in the accompanying solutions booklet.

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1 Given that $2^x = 5$ and $2^y = 20$, find the value of 2^{2x-y} .

- A** $\frac{1}{4}$
- B** $\frac{5}{4}$
- C** $\frac{5}{4}$
- D** -10
- E** $\frac{25}{2}$
- F** 5

2 The inequality $x^2 + bx + 9 > 0$ holds for all real x . Find the complete set of values of b .

A $-3 < b < 3$

B $b < -6$ or $b > 6$

C $b > 6$

D $-6 < b < 6$

E $b \geq 6$

F all real b

- 3 The circle $x^2 + y^2 = r^2$ and the line $3x + 4y = 20$ have exactly one point in common. Find r .
- A $\frac{20}{7}$
 - B 4
 - C 20
 - D 5
 - E $\sqrt{20}$
 - F 2

- 4 Let $I = \int_0^1 (x^2 + 1) dx$ and $J = \int_0^1 (x + 1)^2 dx$. Which of the following is true?
- A** $I < J$ and $J - I = \frac{4}{3}$
- B** $I < J$ and $J - I = \frac{1}{2}$
- C** $I = J$
- D** $I < J$ and $J - I = 1$
- E** $I > J$

- 5 A bacteria population doubles every 3 hours. By what factor does it grow in 24 hours?
- A 16
 - B 48
 - C 64
 - D 128
 - E 8
 - F 256

6 The line $y = 4x + c$ is a tangent to $y = x^2 + 2x + 5$. Find c .

A 4

B -4

C 1

D 5

E 6

F 3

7 Find the exact value of $(\sqrt{5} - 1)^2 + (\sqrt{5} + 1)^2$.

A 12

B 8

C 10

D 6

E $12 + 4\sqrt{5}$

F $4\sqrt{5}$

- 8** A sequence is defined by $u_1 = 1$ and $u_{n+1} = u_n + 2n + 1$. Which of the following is an expression for u_n ?

A $n^2 - n + 1$

B $2n - 1$

C $2^n - 1$

D $n^2 + 1$

E $n(n + 1)/2$

F n^2

- 9** The line $y = mx$ meets the curve $y = x^2 + 9$ at two distinct points. Find the complete set of values of m .
- A** $m < -3$ or $m > 3$
- B** $m < -6$ or $m > 6$
- C** $-6 < m < 6$
- D** $-3 < m < 3$
- E** $m > 6$
- F** all real m

10 Simplify $\sqrt{75} - \sqrt{27} + \sqrt{12}$.

A $2\sqrt{3}$

B $6\sqrt{3}$

C $\sqrt{60}$

D $10\sqrt{3}$

E $4\sqrt{3}$

F $5\sqrt{3}$

11 The circle C has equation $x^2 + y^2 - 4x - 6y + 9 = 0$. Which of the following lines does not meet C ?

A $x = 4$

B $x = 2$

C $x = 0$

D $y = 0$

E $y = 5$

- 12** An arithmetic sequence has $u_3 = 10$ and $u_8 = 25$. Find the smallest value of n for which $u_n > 100$.
- A** 36
 - B** 32
 - C** 34
 - D** 31
 - E** 33
 - F** 35

13 The equation $kx^2 + 4x + k = 0$ has no real roots. Find the complete set of values of k .

A $0 < k < 2$

B $k > 2$

C $k < -2$ or $k > 2$

D $-2 < k < 2$

E $k < -2$

F $-2 < k < 0$ or $0 < k < 2$

- 14** Two similar triangles have corresponding sides of length 6 cm and 9 cm. The smaller triangle has area 20 cm^2 . Find the area of the larger triangle.

- A** 30 cm^2
- B** 67.5 cm^2
- C** 40 cm^2
- D** 45 cm^2
- E** 27 cm^2
- F** 60 cm^2

- 15** The first three terms in the expansion of $(1 + ax)^n$ in ascending powers of x are $1 + 12x + 54x^2$. Find n .
- A** 9
- B** 6
- C** 10
- D** 4
- E** 12
- F** 8

16 Given that $x^3 + px + q$ is divisible by $(x - 2)^2$, find the value of q .

A 16

B 12

C 8

D -8

E -16

F -12

17 Triangle ABC has $AB = 7$, $BC = 8$, $CA = 9$. Find the area of the triangle.

A 28

B $\sqrt{720}$

C $24\sqrt{5}$

D $12\sqrt{5}$

E 30

F $6\sqrt{5}$

- 18** Solve $12\cos^2 x + 6\sin x - 10 = 2$ for $0^\circ < x < 360^\circ$. Which of the following is the complete set of solutions?
- A** $0^\circ, 30^\circ, 150^\circ, 180^\circ$
 - B** $30^\circ, 150^\circ, 210^\circ, 330^\circ$
 - C** $30^\circ, 150^\circ, 270^\circ$
 - D** $30^\circ, 150^\circ$
 - E** $210^\circ, 330^\circ$
 - F** $30^\circ, 150^\circ, 180^\circ$

- 19** Evaluate $\int_1^{43x^2-5} \frac{1}{\sqrt{x}} dx$.
- A** $\frac{146}{5}$
- B** $\frac{186}{5}$
- C** 28
- D** $\frac{86}{5}$
- E** $\frac{126}{5}$
- F** $\frac{136}{5}$

20 How many positive divisors does 360 have?

A 24

B 20

C 18

D 36

E 12

F 16

END OF TEST